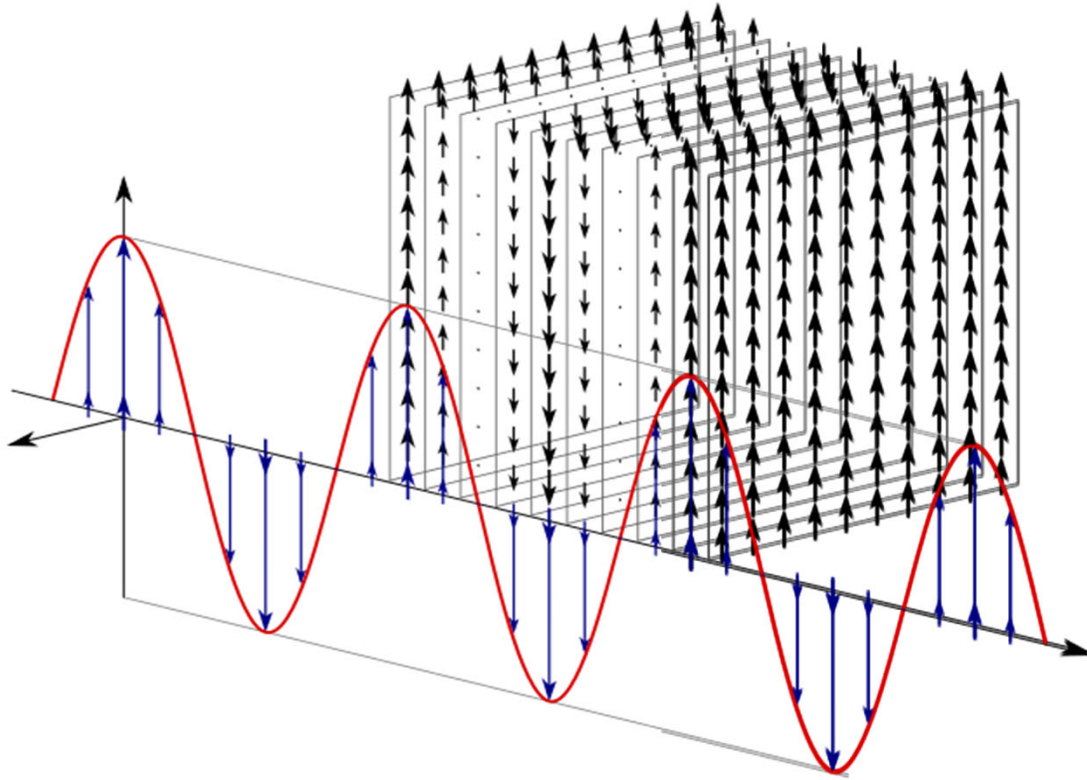


Ph134, Lecture 12

- **Spatial structure of light:**
 - **Plane waves, expanding spherical waves**
 - **Paraxial approximation**
 - **Gaussian beams: Beam waist, Rayleigh range**
 - **(Hermite-Gauss and Laguerre-Gauss modes)**

Plane waves are not beams!



Spatial field patterns that solve the wave equation for monochromatic light:

$$\nabla^2 E^{(+)} - \frac{1}{c^2} \frac{\partial^2 E^{(+)}}{\partial t^2} = 0$$

Separate time and position dependence: $E^{(+)}(\vec{r}, t) = u(\vec{r})T(t)$

$$\nabla^2 u + k^2 u = 0 \quad \frac{d^2 T}{dt^2} + \omega^2 T = 0 \quad \omega = ck = c \frac{2\pi}{\lambda}$$

Helmholtz equation

Let $T(t) = e^{-i\omega t}$.

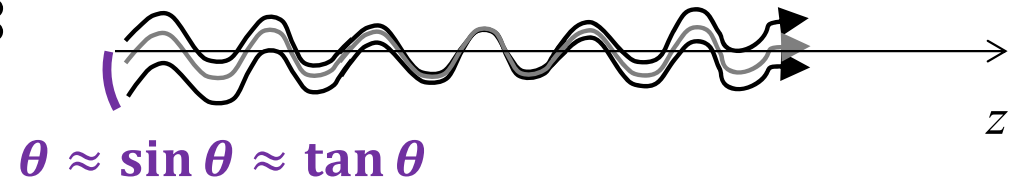
Plane wave: e.g., $u(\vec{r}) = E_0^{(+)} e^{ikz}$ in Cartesian (x, y, z) coords

Spherical wave: e.g., $u(\vec{r}) = A_0^{(+)} \frac{e^{ik\rho}}{\rho}$ in spherical (ρ, θ, φ) coords

Vector solutions that obey full Max. Eq.?

Spherical wave needs multipole expansion, but ρ dependence always $\approx \frac{e^{ik\rho}}{\rho}$ in limit of $\rho \gg \lambda$

Wave equation in the paraxial approximation:



$$E^{(+)}(\vec{r}, t) = u(\vec{r})T(t)$$

$$\nabla^2 u + k^2 u = 0 \quad \frac{d^2 T}{dt^2} + \omega^2 T = 0 \quad \omega = ck$$

Helmholtz equation

Separate wavelength-scale oscillation from slowly-varying amplitude:

$u(\vec{r}) = \psi(\vec{r})e^{ikz}$, with ψ slowly varying in z

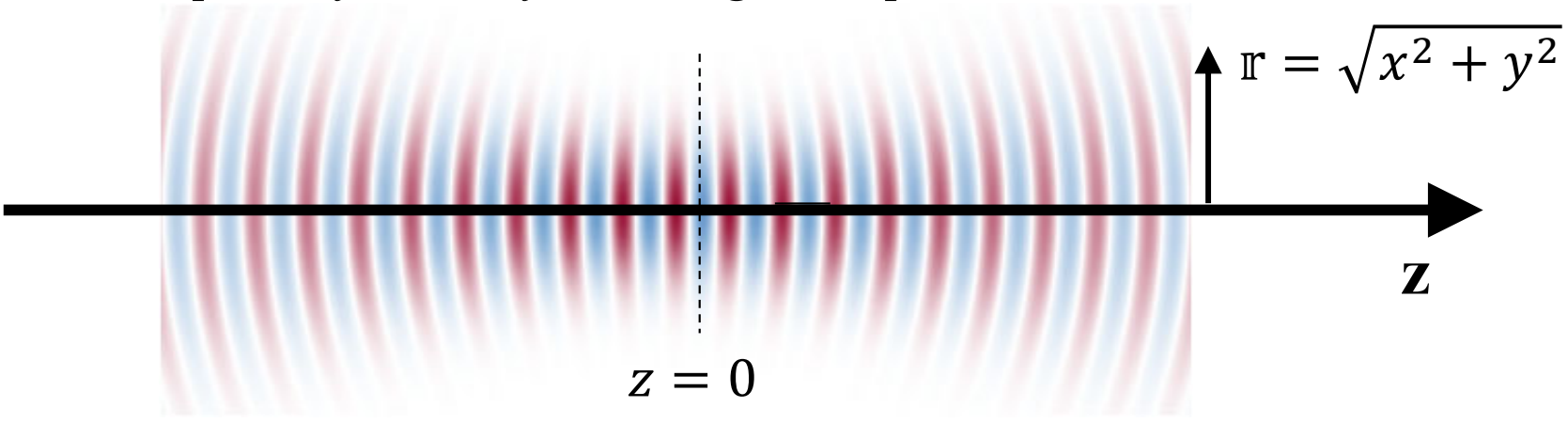
Electric field pattern that (mostly) maintains shape as beam propagates

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + 2ik \frac{\partial \psi}{\partial z} = 0$$

Helmholtz equation in the paraxial approximation

Gaussian beams: cylindrical (r, φ, z) coords

Snapshot of electric field strength and phase:



$$E^{(+)}(r, z, t) = E_0^{(+)} e^{-i\omega t} \frac{w_0}{w(z)} e^{-\frac{r^2}{(w(z))^2}} e^{i\left(\underbrace{kz - \tan^{-1} \frac{z}{z_0} + \frac{kr^2}{2R(z)}}_{\text{Gouy phase shift } \zeta(z)} \right)}$$

beam waist at focus

Gouy phase shift $\zeta(z)$

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_0}\right)^2}$$

beam waist at z

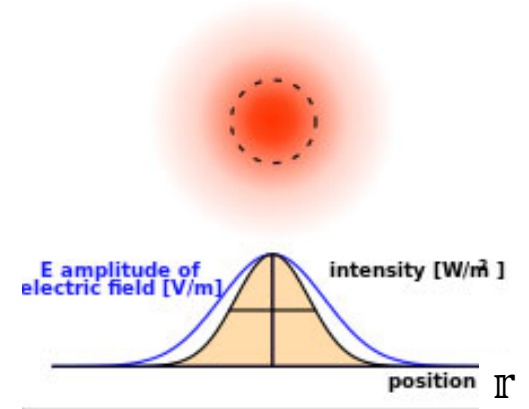
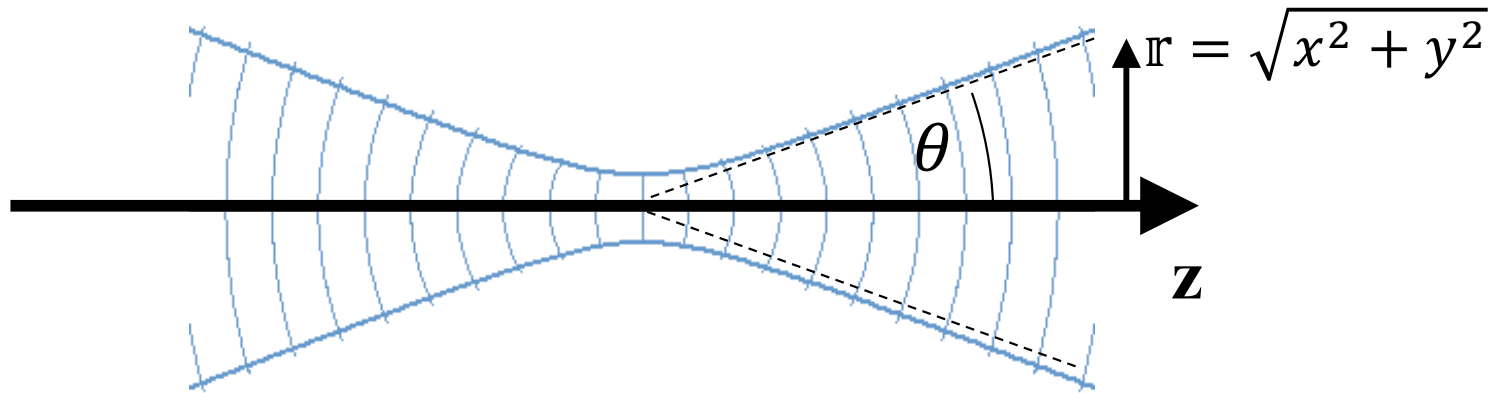
$$R(z) = z \left[1 + \left(\frac{z_0}{z}\right)^2 \right]$$

Rayleigh range

$$z_0 = \frac{\pi w_0^2}{\lambda}$$

Radius of curvature of wavefronts

Properties of Gaussian beams:



$$I(r, z) = I_0 \left(\frac{w_0}{w(z)} \right)^2 e^{-\frac{2r^2}{(w(z))^2}}$$

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_0} \right)^2}$$

Beam divergence $\theta \approx \frac{\lambda}{\pi w_0}$

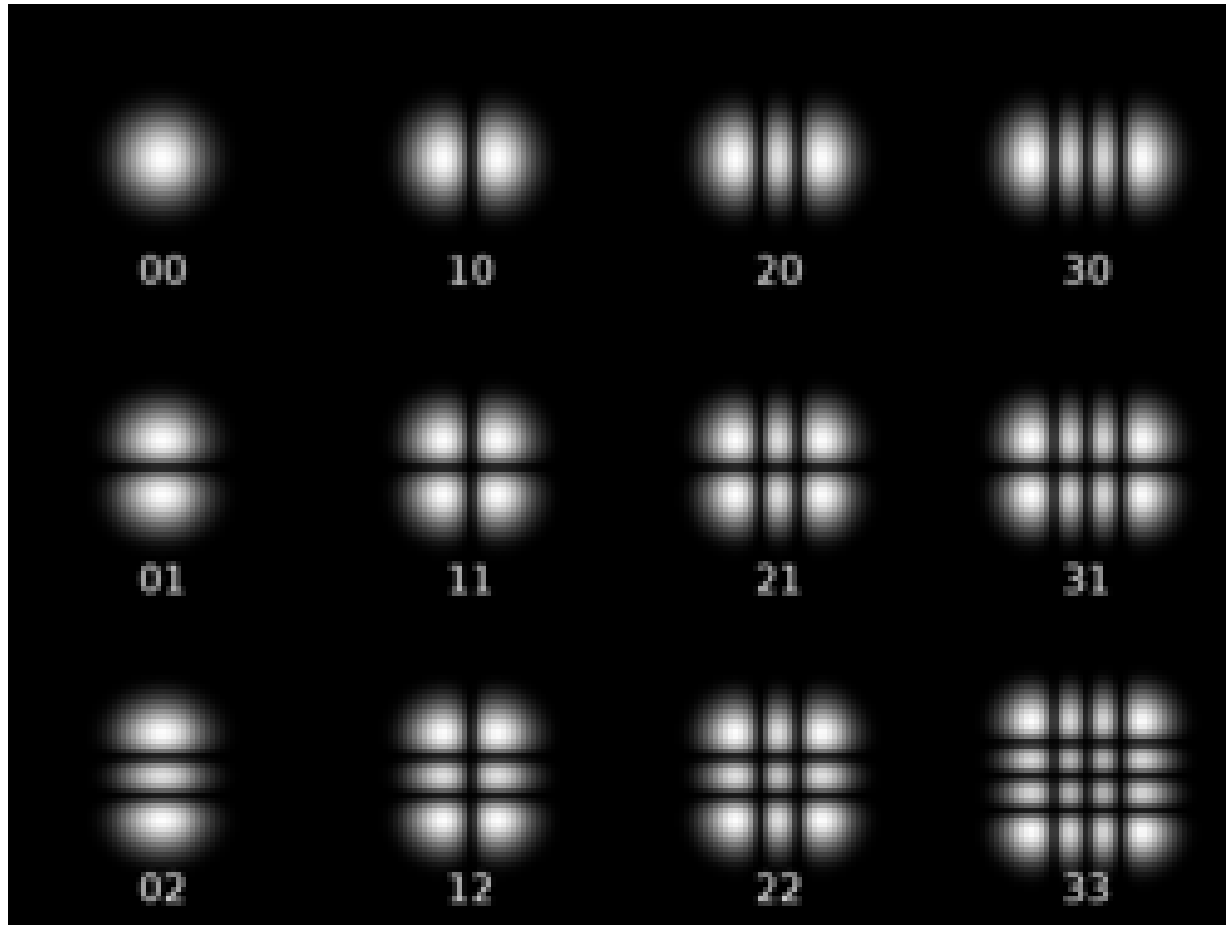
$$z_0 = \frac{\pi w_0^2}{\lambda}$$

Paraxial approx holds if focused spot size $\gtrsim$ wavelength of light

*Gaussian beam $\rightarrow$ curved mirror or lens $\rightarrow$ Gaussian beam
Use ABCD matrices to calculate how optical elements change beam properties*

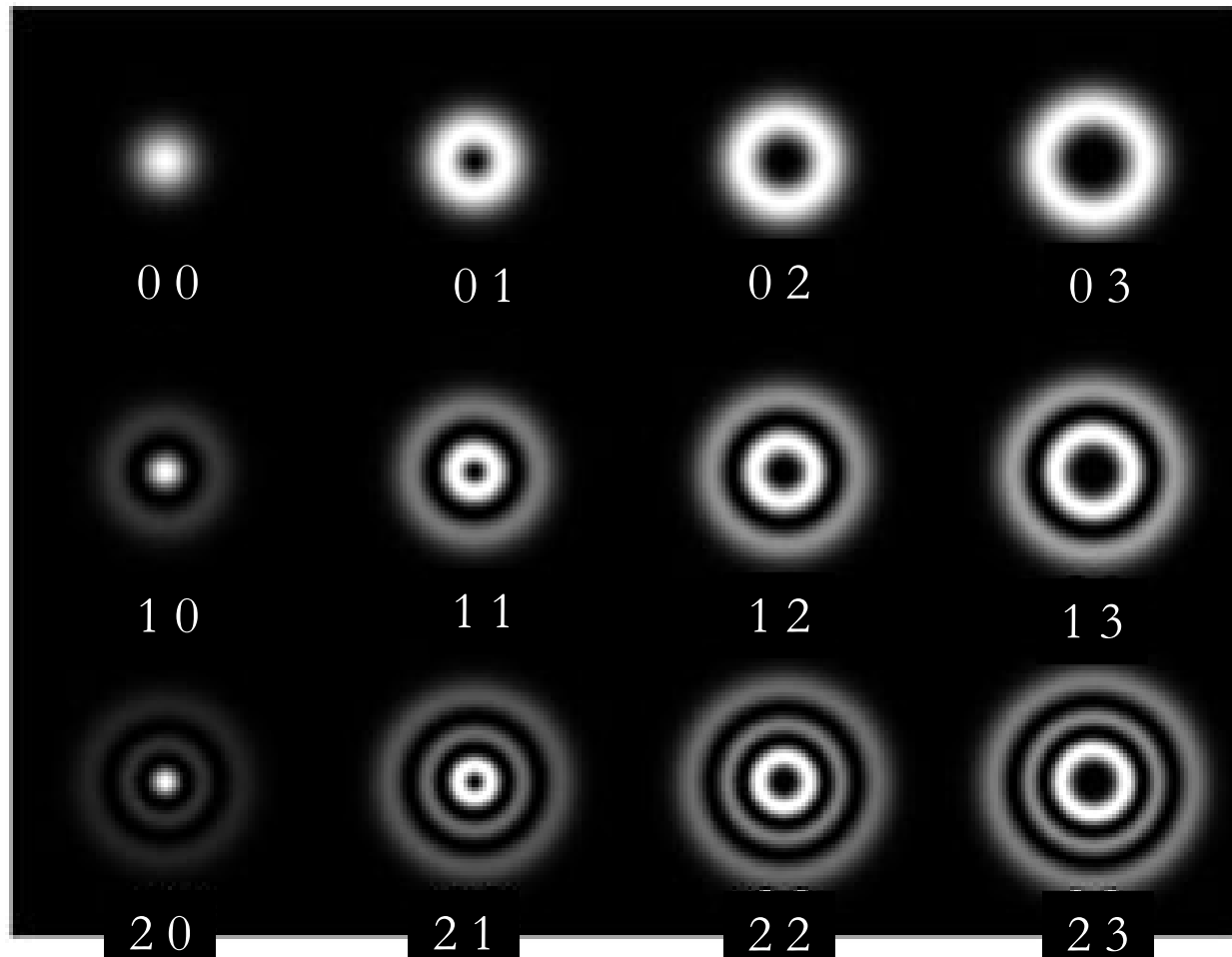
Hermite-Gauss modes:

$$E_{\ell,m}^{(+)}(x,y,z,t) = E_0^{(+)} \sqrt{\frac{1}{2^{\ell+m} \ell! m!}} e^{-i\omega t} \frac{w_0}{w(z)} H_{\ell} \left(\frac{\sqrt{2}x}{w(z)} \right) e^{-\frac{x^2}{(w(z))^2}} \\ \times H_m \left(\frac{\sqrt{2}y}{w(z)} \right) e^{-\frac{y^2}{(w(z))^2}} e^{i \left(kz - (1+m+n) \tan^{-1} \frac{z}{z_0} + \frac{k(x^2+y^2)}{2R(z)} \right)}$$



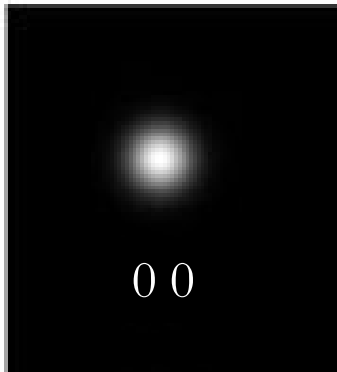
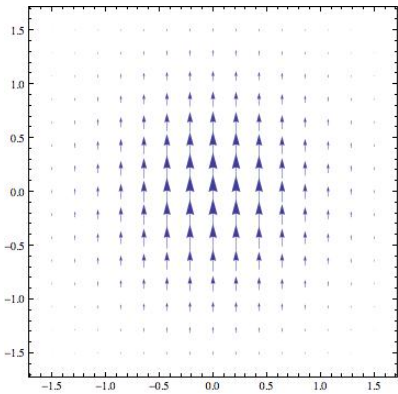
Laguerre-Gauss modes:

$$E_{\ell,m}^{(+)}(r, \varphi, z, t) = E_0^{(+)} \sqrt{\frac{2\ell!}{\pi(\ell + |m|)!}} e^{-i\omega t} \frac{w_0}{w(z)} \left(\frac{\sqrt{2}r}{w(z)} \right)^{|m|} L_{\ell}^{|m|} \left(\frac{2r^2}{(w(z))^2} \right) e^{-\frac{r^2}{(w(z))^2}} \\ \times e^{-im\varphi} e^{i\left(kz - (1+2\ell+|m|) \tan^{-1} \frac{z}{z_R} + \frac{kr^2}{2R(z)}\right)}$$

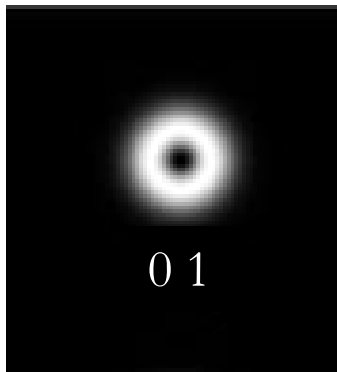
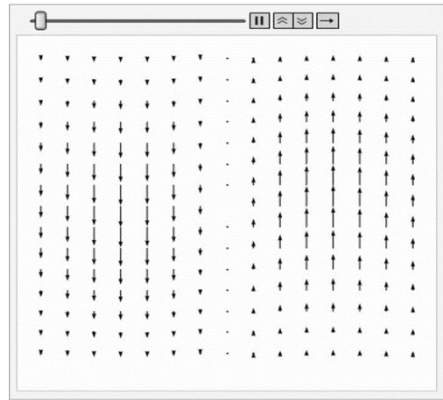


Laguerre-Gauss modes carry orbital angular momentum:

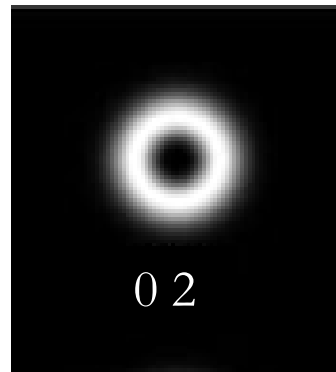
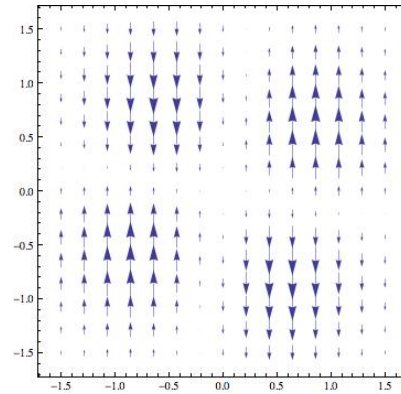
$$E_{\ell,m}^{(+)}(r, \varphi, z, t) = E_0^{(+)} \sqrt{\frac{2\ell!}{\pi(\ell + |m|)!}} e^{-i\omega t} \frac{w_0}{w(z)} \left(\frac{\sqrt{2}r}{w(z)} \right)^{|m|} L_{\ell}^{|m|} \left(\frac{2r^2}{(w(z))^2} \right) e^{-\frac{r^2}{(w(z))^2}} \\ \times e^{-im\varphi} e^{i \left(kz - (1+2\ell+|m|) \tan^{-1} \frac{z}{z_R} + \frac{kr^2}{2R(z)} \right)}$$



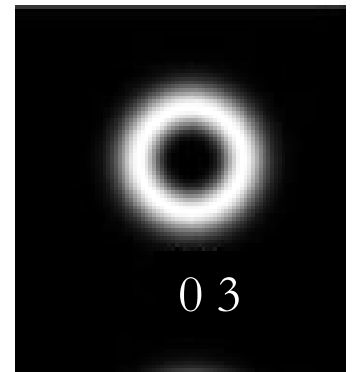
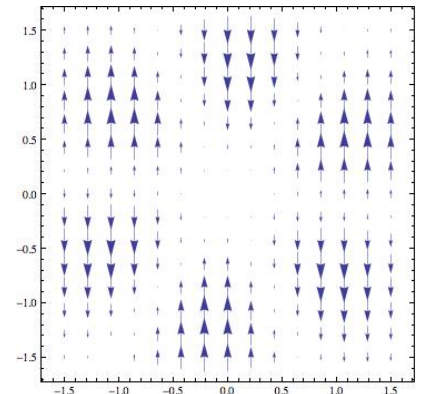
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0 1



0 2



0 3